

```

    path: /var/lib/pgsql/9.6/data/recovery.conf
    block: |
        standby_mode = 'on'
        primary_conninfo = 'host={{ hostvars['host1'].
ansible_default_ipv4.address }} port=5432 user=replica
password={{ replica_password }} application_name=slave01'
        trigger_file = '/tmp/postgresql.trigger.5432'
    mode: 0600
    owner: postgres
    group: postgres
    state: present
    create: yes

- name: Start PostgreSQL server
  systemd:
    name: postgresql-9.6
    state: started

```

The above playbook can be executed using the following command:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/postgresql.yml --tags pgslave -K
```

Testing

You can log in to the master instance, become a Postgres user, and run the following *psql* commands to see the synchronisation state of the setup:

```

# su - postgres
-bash-4.2$ psql -c "select application_name, state, sync_
priority, sync_state from pg_stat_replication;"
 application_name | state   | sync_priority | sync_state
-----+-----+-----+-----
 slave01          | streaming |          1    | sync
(1 row)

-bash-4.2$ psql -x -c "select * from pg_stat_replication;"
-[ RECORD 1 ]-----+-----
pid           | 4054
usesysid      | 16384
username      | replica
application_name | slave01
client_addr   | 192.168.122.113
client_hostname |
client_port   | 53720
backend_start  | 2018-12-12 21:32:07.382766+05:30
backend_xmin   |
state         | streaming
sent_location  | 0/30000060
write_location | 0/30000060
flush_location | 0/30000060

```

```

replay_location | 0/30000060
sync_priority   | 1
sync_state      | sync

```

You can create tables and insert records on the master instance. An example is given below:

```

-bash-4.2$ psql
psql (9.6.11)
Type "help" for help.

postgres=# CREATE TABLE student (name VARCHAR(100));
CREATE TABLE
postgres=# INSERT INTO student VALUES ('Adith');
INSERT 0 1
postgres=# INSERT INTO student VALUES ('Shakthi');
INSERT 0 1
postgres=#

```

You can now verify that the records exist on the slave instance using the following set of commands:

```

[root@host1 ~]# su - postgres
Last login: Wed Dec 12 21:32:02 IST 2018
-bash-4.2$ psql
psql (9.6.11)
Type "help" for help.

postgres=# select * from student;
 name
-----
 Adith
 Shakthi
(2 rows)

```

If you try to insert records on the slave instance, the database will throw a 'read-only transaction' error as shown below:

```

postgres=# INSERT INTO student VALUES ('Foo');
ERROR:  cannot execute INSERT in a read-only transaction
postgres=#

```

You are encouraged to read Chapter 26 of the book 'High Availability, Load Balancing Replication' from the PostgreSQL documentation page at <https://www.postgresql.org/docs/current/high-availability.html>. 

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